Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 24 Question Id : 6911211224 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let the image of the point $P(0,-5,0)$ in the line $\frac{x-1}{2}=\frac{y}{1}=\frac{z+1}{-2}$ be the point R and the image of the point $Q\left(0,\frac{-1}{2},0\right)$ in the line $\frac{x-1}{-1}=\frac{y+9}{4}=\frac{z+1}{1}$ be the point S. Then the square of the area of the parallelogram PQRS is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 25 Question Id : 6911211225 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let $f(x)=\begin{cases}x^3+8\ ; & x<0\ ,\\ x^2-4\ ; & x\geq 0\ ,\end{cases}$ and $g(x)=\begin{cases}(x-8)^{1/3}\ ; & x<0,\\ (x+4)^{1/2}\ ; & x\geq 0\ .\end{cases}$

Then the number of points, where the function *gof* is discontinuous, is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Physics Section A

Section Id :	69112199
Section Number :	3
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69112199
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 26 Question Id : 6911211226 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The percentage error in the calculated volume of a sphere, if there is 2% error in its diameter measurement, is _____.

Options :

6911214166. 1

6911214167. 2

6911214168. 6

6911214169. 8

Question Number : 27 Question Id : 6911211227 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match **List - I** with **List - II**.

	List - I		**List - II**
A.	Boltzmann constant	I.	$[M^{-1} L^3 T^{-2}]$
B.	Stefan's constant	II.	$[M L^2 T^{-1}]$
C.	Planck's constant	III.	$[M L^2 T^{-2} K^{-1}]$
D.	Gravitational constant	IV.	$[M L^0 T^{-3} K^{-4}]$

Choose the **correct** answer from the options given below :

Options :

6911214170. A-I, B-II, C-III, D-IV

6911214171. A-IV, B-III, C-II, D-I

6911214172. A-III, B-IV, C-II, D-I

6911214173. A-II, B-I, C-IV, D-III

Question Number : 28 Question Id : 6911211228 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A solid sphere (A) of mass $5m$ and a spherical shell (B) of mass m, both having same radius, are placed on a rough surface. When a force of same magnitude is applied tangentially at the highest points of A and B, they start rolling without slipping with an acceleration of a_A and a_B, respectively. The ratio of a_A and a_B is _____.

Options :

6911214174. 5 : 21

6911214175. 6 : 10

6911214176. 21 : 25

6911214177. $1:5$

Question Number : 29 Question Id : 6911211229 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A body of mass 1 kg moves along a straight line with a velocity $v = 2x^2$. The work done by the body during displacement from $x = 0$ to 5 m is _____ J.

Options :

6911214178. 0

6911214179. 250

6911214180. 1250

6911214181. 1000

Question Number : 30 Question Id : 6911211230 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A cylinder with adiabatic walls is closed at both ends and is divided into two compartments by a frictionless adiabatic piston. Ideal gas is filled in both (left and right) the compartments at same P, V, T. Heating is started from left side until pressure changes to $^{27}P/_{8}$. If initial volume of each compartment was 9 litres then the final volume in right-hand side compartment is _____ litres. (for this ideal gas $C_P/C_V = 1.5$)

Options :

6911214182. 3

6911214183. 4

6911214184. 14

6911214185. 9

Question Number : 31 Question Id : 6911211231 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

For an electromagnetic wave propagating through vacuum, $\vec{k}$, $\vec{E}$ and ω represent propagation vector, electric field and angular frequency, respectively. The magnetic field associated with this wave is represented by :

Options :

6911214186. $\frac{\vec{E} \times \vec{k}}{\omega}$

6911214187. $\frac{\vec{k} \times \vec{E}}{\omega}$

6911214188. $\omega\left(\vec{E} \times \vec{k}\right)$

6911214189. $\omega\left(\vec{k} \times \vec{E}\right)$

Question Number : 32 Question Id : 6911211232 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Two identical bodies A and B of equal masses have initial velocities $\vec{v}_1 = 4\hat{i}$ m/s and $\vec{v}_2 = 4\hat{j}$ m/s respectively. The body A has acceleration $\vec{a}_1 = 6\hat{i} + 6\hat{j}$ m/s^2 while the acceleration of the other body B is zero. The centre of mass of the two bodies moves in _____ path.

Options :

6911214190. circular

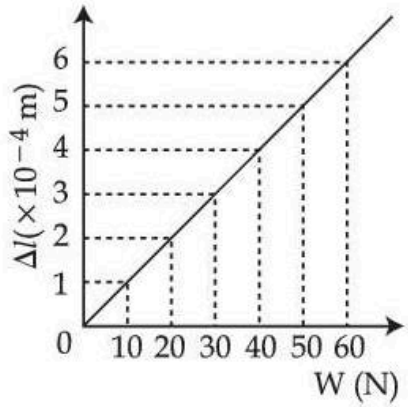
6911214191. parabolic

6911214192. straight line

6911214193. elliptical

Question Number : 33 Question Id : 6911211233 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Figure represents the extension (Δl) of a wire of length 1 meter, suspended from the ceiling of the room at one end with a load W connected to the other end. If the cross-sectional area of the wire is 10^{-5} m^2 then the Young's modulus of the wire is _____ N/m^2.



Options :

6911214194. 1.0×10^{11}

6911214195. 2.0×10^{10}

6911214196. 1.0×10^{10}

6911214197. 2.0×10^{11}

Question Number : 34 Question Id : 6911211234 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A cylindrical vessel of 40 cm radius is completely filled with water and its capacity is 528 dm^3 (dm : decimeter) The vessel is placed on a solid block of exactly same height as vessel. If a small hole is made at 70 cm below the top of water level, then horizontal range of water falling on the ground in the beginning is _____ cm.

Options :

6911214198. $120\sqrt{2}$

6911214199. $140\sqrt{2}$

6911214200. $140\sqrt{3}$

6911214201. $120\sqrt{3}$

Question Number : 35 Question Id : 6911211235 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If 2 mole of an ideal monoatomic gas at temperature T, is mixed with 6 mole of another ideal monoatomic gas at temperature $2T$ then the temperature of mixture is :

Options :

6911214202. $\frac{5}{2}T$

6911214203. $\frac{5}{4}T$

6911214204. $\frac{7}{2}T$

6911214205. $\frac{7}{4}T$

Question Number : 36 Question Id : 6911211236 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A spring stretches by 2 mm when it is loaded with a mass of 200 g. From equilibrium position the mass is further pulled down by 2 mm and released. The frequency associated with the system and maxmimum energy in the spring are _____ Hz and _____ J, respectively.
(Take $g = 10\ m/s^2$)

Options :

6911214206. $\frac{5\sqrt{50}}{\pi}$ and 8×10^{-3}

6911214207. $\frac{5\sqrt{50}}{\pi}$ and 8

6911214208. $10\sqrt{50}$ and 2×10^{-3}

6911214209. $\frac{5\sqrt{50}}{\pi}$ and 16×10^{-3}

Question Number : 37 Question Id : 6911211237 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The electric potential as a function of x, y is given by $V = 5(x^2 - y^2)$ V. The electric field at a point (2, 3) m is _____ V/m.

Options :

6911214210. $\left(-20\hat{i} + 30\hat{j}\right)$

6911214211. $\left(20\hat{i} - 30\hat{j}\right)$

6911214212. $\left(20\hat{i} + 45\hat{j}\right)$

6911214213. $\left(-4\hat{i} + 6\hat{j}\right)$

Question Number : 38 Question Id : 6911211238 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A current of 30 A each flows in opposite directions in two conducting wires, placed parallel to each other at a distance of 8 cm. The magnetic field at the mid point between the two wires is _____ μT.

$(\frac{\mu_o}{4\pi} = 10^{-7}\ N/A^2)$

Options :

6911214214. 30

6911214215. 300

6911214216. 150

6911214217. 0.0

Question Number : 39 Question Id : 6911211239 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A square loop of side 2 cm is placed in a time varying magnetic field with magnitude as $B = 0.4 \sin(300t)$ Tesla. The normal to the plane of loop makes an angle of 60° with the field. The maximum induced emf produced in the loop is _____ mV.

Options :

6911214218. 12

6911214219. 18

6911214220. 21

6911214221. 24

Question Number : 40 Question Id : 6911211240 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A sphere of capacitance 100 pF is charged to a potential of 100 V. Another identical uncharged metal sphere is brought in contact with the charged sphere, then the change in the total energy stored on these spheres, when they touch is $\alpha \times 10^{-7}$ J. The value of α is _____.

(combined capacitance of spheres is 200 pF)

Options :

6911214222. 5

6911214223. $\frac{5}{2}$

6911214224. $\frac{7}{2}$

6911214225. $\frac{9}{2}$

Question Number : 41 Question Id : 6911211241 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The energy released if hydrogen atoms are combined to form $^{4}_{2}He$ is _____ MeV.

(Take binding energies per nucleon of $^{2}_{1}H$ and $^{4}_{2}He$ as 1.1 MeV and 7.2 MeV, respectively)

Options :

6911214226. 6.1

6911214227. 24.4

6911214228. 26.6

6911214229. 5

Question Number : 42 Question Id : 6911211242 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Angle of minimum deviation is equal to the half of the angle of prism in an equilateral prism. The refractive index of the prism is _____.

Options :

6911214230. 1.5

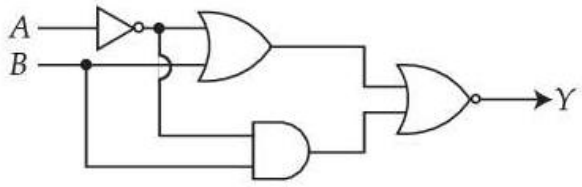
6911214231. $\sqrt{3}$

6911214232. $\sqrt{2}$

6911214233. 1.65

Question Number : 43 Question Id : 6911211243 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Refer to the logic circuit given below. For two inputs $(A=1, B=1)$ and $(A=0, B=1)$, output (Y) will be _____.



Options :

6911214234. 1, 0 respectively

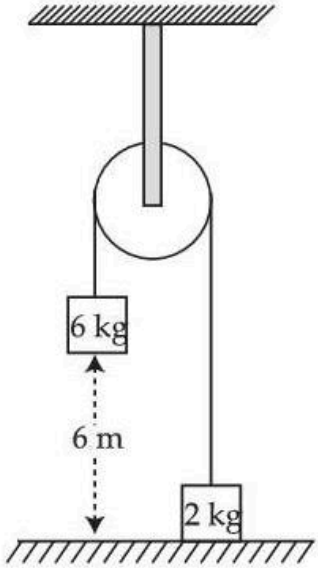
6911214235. 0, 1 respectively

6911214236. 0, 0 respectively

6911214237. 1, 1 respectively

Question Number : 44 Question Id : 6911211244 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The velocity at which 6 kg mass (shown in figure) strikes the ground when it is released from a height of 6 m above the ground is _____ m/s. Assume pulley is massless and string is light and inextensible. (Take $g = 10\ m/s^2$)



Options :

6911214238. 7.74

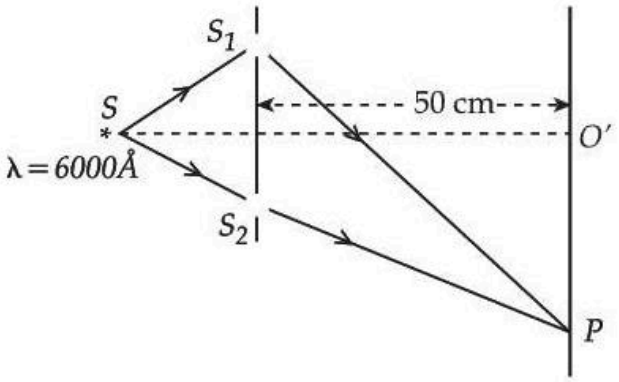
6911214239. 7.20

6911214240. 6.55

6911214241. 4.50

Question Number : 45 Question Id : 6911211245 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

In a Young double slit experiment, the wavelength of incident light is 6000 Å, the separation between slits S_1 and S_2 is 5 cm and the distance between slits plane and screen is 50 cm, as shown in the figure below. If the resultant intensity at P is equal to the intensity due to individual slits, the path difference between interfering waves is _____ Å.



Options :

6911214242. 4000

6911214243. 3000

6911214244. 2000

6911214245. 1000

Physics Section B

Section Id :	691121100
Section Number :	4
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	691121100
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 46 Question Id : 6911211246 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A block takes t time to slide down a plane inclined at 45° to the horizontal. If the surface is made smooth (frictionless), the block takes time $\frac{t}{2}$ to slide down the plane. The coefficient of friction between the block and the inclined plane is $\left(\frac{\alpha}{100}\right)$. The value of α is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Possible Answers :

1

Question Number : 47 Question Id : 6911211247 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

The de Broglie wavelength for an electron accelerated through the potential difference of V_1 volt is λ_1. When the potential difference is changed to V_2 volt, the associated de Broglie wavelength is increased by 50%. If $(V_1/V_2)=(9/\alpha)$, then the value of α is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1

Question Number : 48 Question Id : 6911211248 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A moving coil of galvanometer when shunted with 2 Ω resistance gives a full scale deflection for a current of 500 mA. When a resistance of 470 Ω is connected in series it gives a full scale deflection for 10 V potential applied on it. The value of resistance of galvanometer coil is _____ Ω.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1

Question Number : 49 Question Id : 6911211249 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Two cells of emfs 1 V and 2 V and internal resistance 2 Ω and 1 Ω, respectively connected in parallel, gave a current of 1 A through an external resistance. If the polarity of one cell is reversed, then value of current through the external resistance will be $\frac{\alpha}{5}$ A. The value of α is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1

Question Number : 50 Question Id : 6911211250 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A concave mirror of focal length 10 cm forms an image which is double the size of object when the object is placed at two different positions. The distance between the two positions of the object is _____ cm.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :